

D PROJECT DESCRIPTION

D.1. PROPOSAL OBJECTIVES

Brain Machine Interfaces (BMIs) are an emerging technology whose purpose is to allow amputees and spinal cord injury patients to control a prosthetic limb using signals derived from the brain. BMIs are vital to investigate because, in addition to their practical applications, their study leads to valuable research tools and fundamental new theories in neuroscience. Likewise, the research in this proposal will not only lead to improved designs for BMI instrumentation (especially wireless and implantable systems) but in a broader sense will also lead to modeling methods that will be useful to neuroscience researchers studying motor learning or cortical adaptation as well as to graduate students training in neural engineering.

The proposed research will seek to develop tools for answering the important question of ***how can neural adaptation be exploited in designing BMI instrumentation***. This question will be addressed by developing a system that models the interaction between an adaptive neural motor control loop, BMI instrumentation, and a prosthetic limb; a key, unique feature will be real-time visual and proprioceptive feedback that will be used for adaptively training synaptic weights in a virtual brain. In doing so, it will become possible to model closed-loop brain adaptation in primate motor control, thus leading to a better understanding of how this phenomenon might be exploited in optimizing BMI instrumentation for various applications. This knowledge will be of particular benefit for designers of implantable and portable BMI instrumentation since optimized designs will result in units that are smaller, consume less power, and dissipate less heat.

While Research Aims 1 and 2 will individually represent important methodological advances in BMI research, their combination and analysis in Research Aim 3 will yield vital hypotheses for guiding BMI design. A set of Educational Aims will complement the Research Aims by adapting the proposed research tools into educational toolkits for neural engineering students, and by developing an environment at Temple University where students at various levels can make meaningful contributions to neural engineering research. The specific aims of this proposal are:

Research Aim 1: Design and implement a simulation platform for modeling adaptive motor control of a human limb in three-space. The proposed simulator will comprise an *in silico* model of the human motor cortex that controls a modeled human arm. Trajectory correction signals will be derived from an *in silico* cerebellum model whose synaptic weights are dynamically tuned via multimodal sensory feedback from the modeled arm. The majority of the model's components will be based on pre-existing, biologically plausible elements, with multiple elements having been identified as candidates for each model component. The simulator will be calibrated for accuracy against pre-existing data from two-dimensional Fitts' Law reaching tasks.

Research Aim 2: Design and implement a programmable logic-based testbed platform for realizing entire families of BMI data acquisition subsystems and systematically manipulating their parameters. The system will simulate the complete BMI signal processing pathway for a 50 x 50 array of cortical electrodes using networks of massively parallel field-programmable gate arrays (FPGAs). Software-based plug-in modules will emulate different algorithms for the various stages in the BMI signal processing pathway. The system will log the complete history of each spike to quantify how and where information is lost or altered.

Research Aim 3: Quantify how BMI performance can be expected to degrade in response to errors in spike detection, spike sorting, and wireless neural data transmission. The tools developed in Research Aims 1 and 2 will be combined and used to investigate adaptation in BMI control. In particular, the PI will quantify how BMI performance varies as a function of error rates in the various spike processing blocks of the BMI signal pathway. The PI will use

these results to identify instrumentation architectures that are optimally suited for wireless and implantable BMIs.

Education Aim 1: Mentor and involve students at the graduate, undergraduate and high school levels as key contributors to the research effort. A main goal of this proposal is to develop the infrastructure for conducting neural engineering at Temple University. It will therefore be of paramount importance to train students at the Doctoral, Masters, and undergraduate levels to conduct research in this area. To promote the development of future neural engineers, one outstanding high school student per year will also be invited to conduct research in the PI's laboratory.

Education Aim 2: Develop a graduate level neural engineering course for the Temple Bioengineering program. Another key element of establishing a neural engineering research track at Temple University will be to create a graduate level course for training prospective research assistants. The course will introduce students to key areas in the field with an emphasis on the theoretical bases of neural engineering research and some of the practical tools (hardware, software, and mathematical) necessary for its execution. It is anticipated that this course will be migrated into Temple's forthcoming Department of Bioengineering where it is projected that the PI will lead the research efforts of the neural engineering group.

Education Aim 3: Develop and disseminate hands-on laboratory-style toolkits based on Research Aims 1 and 2 for training graduate-level neural engineering students. This Aim will help fill a significant deficit in the neural engineering pedagogy by providing graduate students with toolkits that will allow practical experimentation in signal processing, data analysis, statistics, modeling, and motor control. The toolkits will be free and open source, and will be disseminated primarily via presentations given at engineering education conferences. The toolkits' value will be measured via anonymous surveys.

Education Aim 4: Initiate a program to introduce upper level students at the neighboring High School of Engineering and Science to bioengineering and biomedical instrumentation. As an effort to develop an awareness of bioengineering in potential future engineers, an after school program will be established with Temple's neighboring high school. Participating students will visit the PI's lab once a week for a semester and will engage in a series of experiential laboratory exercises designed to expose them to the basics of biomedical instrumentation. This program is expected to significantly impact minority students as the student body is 81% African-American, 8% Asian, 5% Latino, and 5% Caucasian. The program will be evaluated once per semester using anonymous surveys.

D.2. INTELLECTUAL MERIT & BROADER IMPACT

The work in this proposal is driven by two major integrated goals. The first is to develop a new methodology for studying Brain Machine Interfaces, specifically a model that will give insight as to how the plasticity of the brain can be exploited to develop more efficient BMI instrumentation. The second is to expand the tools for conducting research and education in neural engineering at Temple University's College of Engineering. These goals both contribute to the intellectual merit and broader impacts of this proposal.

The *intellectual merit* of this work is that it will create the means for optimizing BMI instrumentation and for determining the minimum design constraints for different classes of BMIs. This in turn will lead to practical solutions for implantable and wireless BMI instrumentation. The proposed work will be an important advance in this field since the lack of fully-implantable BMIs is a major limitation that is preventing this promising technology from becoming a viable therapy for patients of spinal cord injuries and amputations. Further contributing to the merit of this proposal is that Research Aims 1 and 2 individually represent

research tools that will be valuable to many investigators studying neural engineering in general and Brain Machine Interfaces in particular. The Education Aims also positively impact the merit of this proposal by developing new means for educating and mentoring a core group of students to participate in the research.

The **broader impact** of this work is that it will advance techniques for modeling functional ensembles of cortical neurons while also bringing the pedagogical tools for conducting neural engineering education and research to Temple University. The coursework that will be developed, in conjunction with the development of a core group of research-active students, will serve as the basis for establishing a viable neural engineering research track at Temple University. This work will also impact neural engineering education beyond Temple University through the development and dissemination of a framework for flexible, open-source teaching lab kits.

D.3. RESEARCH PLAN

D.3.1. Introduction and Relation to Present State of Knowledge

Limb amputations affect some 1.9 million in the United States alone; spinal cord injuries affect another 400,000 individuals. Brain Machine Interfaces (BMIs) are hardware and software systems designed to aid such patients by allowing them subconscious volitional control over prosthetic limbs. Although it has been nearly 40 years since Fetz first discovered that primates could operantly condition cortical neurons [1], BMI research has yielded few practical results until recently, owing to a lack of sufficiently small and powerful signal processing computers. However in the past five years, investigators have successfully demonstrated primates controlling multi-dimensional prostheses purely under brain control [2-5], and initial trials in human subjects are under way [6]. Although BMIs can be driven by scalp (EEG) [7,8] or cortical surface recordings (ECoG) [9,10], it is held that BMIs based on implanted cortical electrodes will result in the greatest level of dexterity in controlling a prosthetic limb [11,12]. The proposed research deals exclusively with this subset of BMIs.

BMI data acquisition systems are typically comprised of a number of subsystems that include spike detection, spike sorting, spike count binning, and signal reconstruction. The physical bulk of this instrumentation and the fact that subjects must be tethered to it places fundamental limitations on experimental protocols and limits the practical viability of this technology [11,13]. Furthermore, the computationally and bandwidth intensive nature of the requisite signal processing is not well suited to being miniaturized and made wireless. An extensive literature exists investigating novel, optimized algorithms for implementing each of the BMI instrumentation subsystems. Some studies are motivated by the goal of improving performance, regardless of the computational requirements, while others are designed to optimize performance by using minimalist algorithms targeted for implantable and/or wireless applications. For example, spike detection algorithms have been studied that range from computationally trivial threshold detectors [12], to energy-based detectors [14-16], to template-based detectors [17], to computationally demanding wavelet-based detectors [18-20]. Myriad spike sorting algorithms have been tested [15,21-23]; the necessity of sorting prior to reconstruction has also been investigated [2,24,25]. Reconstruction/decoding algorithms have been assessed to determine whether linear or non-linear strategies are better suited for BMI applications [26,27]. And finally, a number of designs for putative wireless BMIs have been investigated centering largely around methods for reducing bandwidth [28-32], latency [33], and size, weight, and power [34-40].

In the studies cited above, the tacit assumption is almost always that “more information is better.” For example, if two spike detectors requiring equal computations per sample are compared and one is found to detect 10% more spikes at a given false alarm rate, then it would

be considered superior. However such logic neglects the fact that, by definition, BMI instrumentation operates on signals generated under closed-loop feedback conditions by brain cells that are constantly adapting their performance to achieve optimal control of a BMI task. Studies in which improvements in BMI performance are accompanied by changes in neuronal tuning properties clearly indicate that the brain adapts itself to maximize its use of the BMI data acquisition system [4]. Therefore, the question engineers ask should not be “how to detect the most spikes” but rather “what is the minimum number of spikes that must be correctly detected without compromising the brain’s ability to adequately control the prosthesis.” Similar questions may be posed relating to how accurate spike sorting must be, how many dropped wireless data packets may be tolerated, and whether it is better to invest system resources for increasing the channel count or for improving spike processing on the existing channels. *These issues will not be properly addressed if the impact of the brain’s plasticity is neglected.* While the definitive procedure for addressing this issue would be to implement a BMI protocol in a large number of animal subjects using myriad combinations of subsystem configurations, this option is patently unrealistic due to ethical, financial, and time considerations. Instead, the model-based approach proposed here will be more feasible.

An overview of the proposed research is shown in Figure 1. At the conclusion of the proposed work, the PI will have developed (*Research Aim 1*) a real-time simulation system that models adaptive plasticity in the cerebellum as pertaining to motor control, (*Research Aim 2*) a programmable logic-based system for performing BMI signal processing on approximately ten times the largest number of channels used in practice at present, and (*Research Aim 3*) a series of testable hypotheses regarding how various tradeoffs in BMI design can be expected to impact performance in motor control tasks.

D.3.2. Research Aim 1

The purpose of Research Aim 1 is to create a new model of motor control in human reaching and tracking featuring a biologically plausible adaptive cerebellum that learns to anticipate limb dynamics and kinematics (see Figure 1). Target trajectories will be subdivided into series of intermediate “desired positions” that will be input into a controller. The controller will convert the desired position into joint angles [41], with trajectory correction feedback being supplied by a

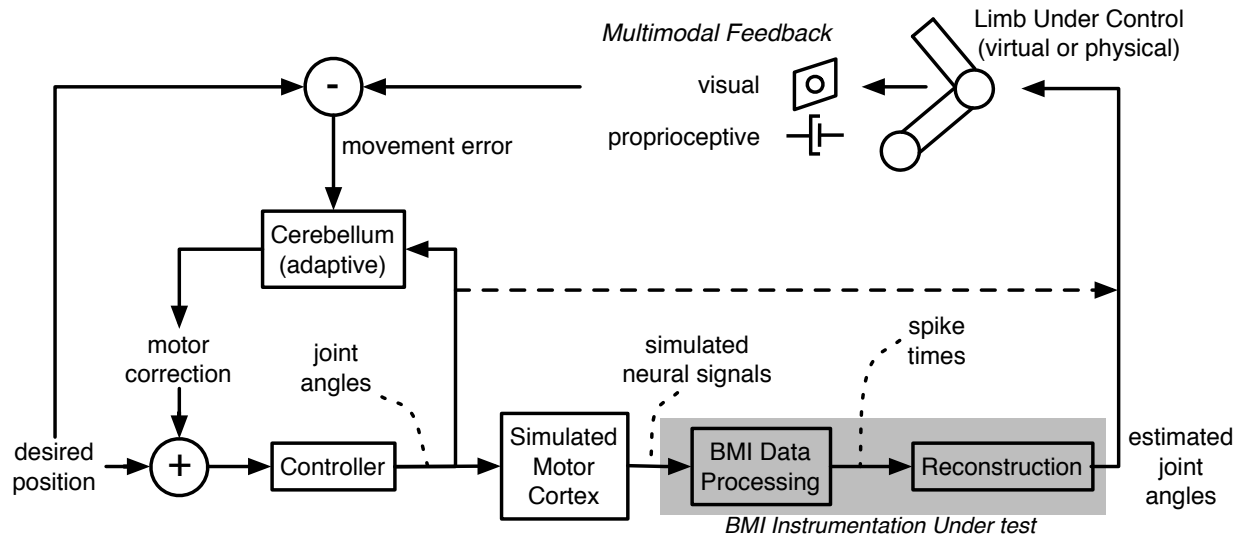


Figure 1: Overview of the entire proposed system. The feedback loop comprising the controller, cerebellum, simulated motor cortex, and robot arm are the subject of Research Aim 1. A testbed platform for rapidly evaluating different configurations of BMI instrumentation is the subject of Research Aim 2 (grey box). The model will be used to evaluate various BMI configurations in Research Aim 3.

virtual cerebellum. The joint angles will be fed to a “limb under control” (following the dashed line in Figure 1) which will be either a 3D air-muscle actuated robotic arm or a virtual 3D upper limb model (developed at Case Western Reserve University) [42]. Multimodal feedback signals (visual and proprioceptive) will be fed back to the virtual cerebellum where they will be compared against the desired arm position in order to produce an error signal for adaptively training the cerebellum. In this manner, it is expected that the cerebellum will adapt over time by tailoring its motor control correction signals to account for the learned static and dynamic properties of the arm, ultimately leading to error minimization in pointing and tracking tasks. The model will include an optional motor cortex simulator based on the NEURON simulation platform that will encode joint angles along with confounding information in order to produce analog extracellular neural recordings containing action potentials tuned to various kinematic inputs.

The proposed system is based on a biologically plausible mathematical treatment [43] of the adaptive filter interpretation [44] of the Marr-Albus cerebellum model. Briefly, the desired motor angles are generated by the controller using the inverse kinematics equations of the robot arm. These signals are used as input both to the robot arm and to the cerebellum via the mossy fiber pathway (see Figure 2). The mossy fibers drive an array of granule cells which effectively enforce a radial basis function (RBF). Each RBF element produces an output signal (carried by a corresponding parallel fiber) whose magnitude is determined by the physical proximity in motor-angle space of the inputs to the element’s center. The parallel fibers all synapse onto a single purkinje cell whose output encodes the motor angle error correction signal that is fed back into the controller. The synaptic weights at the parallel fiber / purkinje cell synapse are iteratively adapted using a gradient descent learning rule that is modulated by the position error introduced to the cerebellum via a climbing fiber. One important advantage of this recurrent architecture is that the teaching signal that trains the cerebellum is based on position error (which is directly visually observable) as opposed to estimates of motor angle error (which must be estimated) as would be required with a feed-forward architecture.

The proposed system’s modular design will allow the evaluation of alternate cerebellar architectures, which have been well addressed in the literature [45-47]. Particular emphasis will be placed on evaluating models whose mathematical stability has already been demonstrated in the context of robotic motor control. For example, Assad et. al. have verified the stability of a cerebellar model that includes an ascending branch (granule cell axon) that synapses onto the parallel fibers. Therefore, whereas the basic Porrill model includes a single plasticity site, the Assad model includes both long term potentiation at the ascending branch / parallel fiber synapse and long term depression at the parallel fiber / purkinje cell synapse [48]. The PI will be assisted in the treatment of cerebellar models by Dr. Diana Woodruff-Pak at Temple University who has studied the cerebellum throughout her career as it pertains to memory, learning, and aging [49-51].

The final component of the proposed simulator is a novel software module that will encode the desired joint angles into realistic spike trains and/or analog extracellular electrode traces. Although the free open-source neural simulator NEURON is capable of simulating neuronal networks of arbitrary complexity, large populations of neurons must still be constructed manually by the user. A new tool is under development that

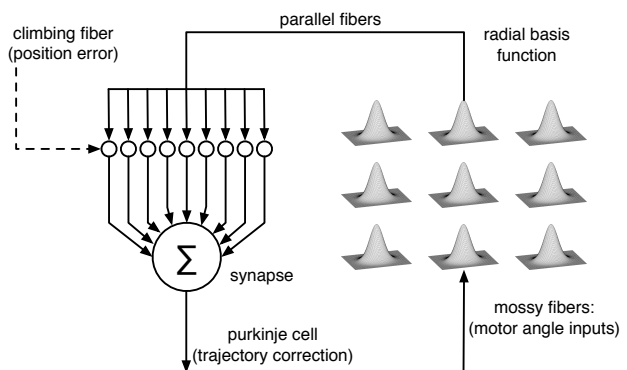


Figure 2: Internal adaptation mechanism of the virtual cerebellum.

will automatically generate neuronal architectures for three dimensional cortical cell populations with biologically plausible distributions of neurons, preferred directions, and synaptic connections [52,53]. Fractal terrain maps are a class of randomization algorithms that generate ‘terrains’ whose spatial distribution of elevations can be defined by the user [54]. These will be used to generate spatial distributions of neuron type and preferred direction that are consistent with published empirical data for various cortical regions [53,55]. The resulting distribution maps will be layered to form a virtual scaffold in which cells are embedded along with their morphological definitions. By this method, a structure of hundreds to thousands of various classes of neurons may be generated pre-runtime from an anatomical definition and simulated in the NEURON environment. NEURON supports plastic synapses, and therefore the models produced by this method can be expected to adapt in response to closed-loop feedback. Over 1000 NEURON-compliant cell models from dozens of labs worldwide have been developed and verified and are freely available online; these will integrate seamlessly into the proposed simulation module [56,57]. The infrastructure of this component is already under active development in the PI’s lab [58].

The system will be validated against published data describing the time course of motor learning and the fundamental limits of accuracy versus speed in a repetitive tracking task. During the validation phase, the Controller output will be wired directly to the limb model (see dashed line, Figure 1). Flament et al demonstrated the time course of learning in an elbow movement task that is well suited for adaptation to the proposed system [59]. The authors measured the rate of improvement and the time to plateau for movement metrics including acceleration and deceleration times, movement durations, and peak acceleration, deceleration, and velocities. Model performance will also be evaluated against both one and two dimensional versions of Fitts’ Law, which concerns the accuracies of movements that have already been learned. In the one dimensional version of Fitts’ Law [60], movement duration is measured as the hand moves back and forth between two targets whose size and separation can be varied. Movement time (T) increases with target separation distance (D) and decreases with target width (W) according to: $T = a + b \log_2(D/W + 1)$, where a and b are empirical constants. A two dimensional version of Fitts’ Law quantifies movement time as subjects draw ellipses, passing through four targets placed at the ends of the ellipse’s major and minor axes [61]. In this case, movement time will be expected to exhibit the same tradeoffs between movement distance and target width as in the one dimensional case. Both versions of Fitts’ Law are well suited for testing the proposed system; in collaboration with Dr. Woodruff-Pak, the model parameters will be varied until the relative time courses in [59] and the Fitts characteristic log-law behavior are observed.

A simplified proof-of-concept version of the proposed system has already been built and tested in the PI’s lab. A two degree-of-freedom robot arm actuated by servo motors was controlled in real-time using visual feedback of the arm’s position to train an adaptive cerebellum similar to the one proposed in the present work [62]. The system was implemented using a radial basis function with 16 equally spaced center elements that spanned the range of shoulder and elbow joint angles necessary for reaching

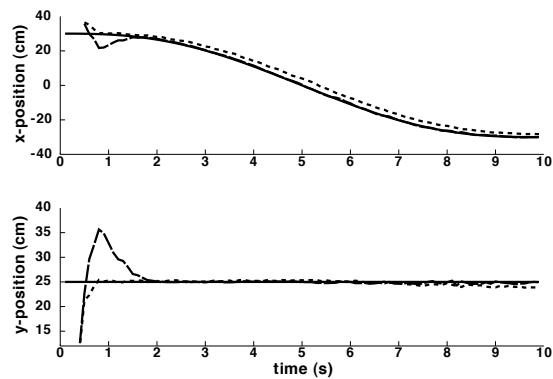


Figure 3: Data from prototype of proposed adaptive reaching model. Trajectory was a minimum-jerk right-to-left horizontal sweep. Solid trace: Target trajectory; Dashed trace: Without adaptive cerebellar feedback; Dotted trace: With adaptive cerebellar feedback.

to all points in front of the arm. All software was written in C++ and was run on a single standard PC. A series of target tracking tasks were simulated, including parabolic and straight line trajectories. Figure 3 shows results from a typical straight line (minimum jerk) horizontal trajectory. Trajectory correction from the cerebellum was shown to typically improve target tracking by a factor of three; in this case, rms error was reduced in the x-direction from 2.0 to 0.6 cm and in the y-direction from 0.6 to 0.2 cm. The essential infrastructure of the system proposed in Research Aim 1 has therefore already been established.

The use of real-time computational cerebellar models to control simulated muscle groups has already been validated by several groups as a practical and viable approach for studying motor learning. Porrill et al have used this paradigm to study adaptation in the vestibulo-ocular reflex (VOR), where the cerebellum is known to play a central role in planning eye movements that minimize retinal image slip as the head turns. Porrill used an adaptive filter implementation of the Marr-Albus cerebellum model [43] to supply corrective movements to a motor controller driving a model of the oculomotor plant [63]. Their model uncovered the role of brainstem in the VOR, where adaptation had long been demonstrated but the purpose of which had been unknown [64]. The model definitively demonstrated that cerebellar plasticity alone would only explain VOR stability up to 2.5Hz; a second plasticity site in a brainstem module was necessary before the model would exhibit stability up to 25Hz as is observed in healthy human adults. The authors have since modified their model to control a physical oculomotor plant comprised of three concentrically mounted rings (roll, pitch, and yaw) and a camera as a means of studying VOR adaptation in a physical embodiment [65].

Adaptive cerebellar modules have also been demonstrated in the context of adaptable biomorphic robots that are designed to general specifications but then adaptively learn how their specific dynamics interact with the environment. Assad et al used a biologically plausible adaptive cerebellar module to drive both a simulated arm [48] and an air muscle-actuated robot arm [66] to throw a ball at a small target. As in the proposed work, each joint was actuated by a pair of antagonistic air muscles. In both cases, a small number of Purkinje Cells interconnected with a few thousand Parallel Fibers were sufficient to map the arm's state-space and to achieve the desired performance within a few hundred trials. The use of air muscles, both here and in the proposed work, is appropriate because they have been demonstrated to possess dynamic qualities similar to human muscle, especially in terms of compliance and force-length properties [67]. Although modifications such as hydraulic dampers and artificial tendons can dampen air-muscles to make them more human-like [68], these are not necessary to observe biologically consistent plasticity in the modeled cerebellum [66].

D.3.3. Research Aim 2

A software reconfigurable BMI (srBMI) based on programmable logic devices will be developed for acquiring and processing up to 2,500 analog extracellular electrode signals (a 50 x 50 electrode array). The system will be capable of interfacing with the simulated motor cortex of Research Aim 1 in order to facilitate the comparison of various BMI instrumentation configurations as specified in Research Aim 3. Owing to the large number of proposed channels and the highly parallel nature of the processing, new approaches will have to be developed for distributing the design over multiple inter-networked programmable logic devices. Beyond its value for evaluating BMIs, the srBMI will establish a new direction for system topologies in BMI instrumentation. The majority of today's BMI instrumentation is built around decades-old DSP technology that is ill-suited for expansion to even hundreds of channels. This Aim will determine how programmable logic devices can be used to create next generation instrumentation that is better suited for expansion to the thousands of channels that researchers will need in the coming decades.

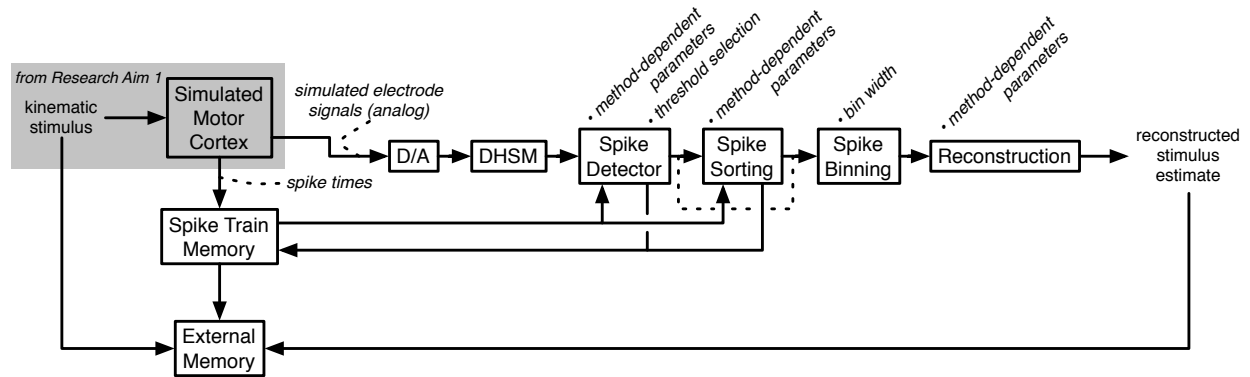


Figure 4: Block diagram of waveform generator and software reconfigurable BMI (srBMI). The spike processing blocks in the srBMI can be implemented with various algorithms. Optional configurations, such as bypassing Spike Sorting, can be specified by the user. The text at top represents examples of user-definable parameters for each block.

A unique feature of the srBMI is that it will quantify the performance statistics of the various BMI instrumentation subsystems in real-time. This will be a valuable tool for investigating how BMI instrumentation affects the information content of the neural signals at every step. The srBMI will be software-coded in either Verilog or C and implemented in an array of interconnected high-performance programmable-logic devices such as the Xilinx (San Jose, CA) Virtex 4 family. The code will consist of a *modular framework* into which different implementations of the various signal processing blocks can be plugged. Each processing block will have a well-defined standard interface; any algorithm implementation satisfying the standard will be swappable into the srBMI. The system will be controlled via a Labview interface that will allow the user to manipulate (either manually or automatically) all of the system parameters for each of the processing blocks.

A block diagram for the srBMI is shown in Figure 4. The srBMI will process the artificial extracellular electrode signals created by the Simulated Motor Cortex described in Research Aim 1. The simulated cortex will also supply parallel signals indicating firing times of modeled neurons (see “*spike times*” in Figure 4). This will provide a robust reference against which to evaluate the spike processing blocks. The srBMI will comprise four such blocks for performing classical BMI signal processing tasks: spike detection, spike sorting, spike binning, and signal reconstruction. In addition to these blocks, the srBMI will use pre-existing digitizing headstage modules (DHSM) to amplify, filter, and digitize the synthetic analog electrode waveforms before they are processed [30].

Spike processing statistics will be collected using a centralized Spike Train Memory. This block will contain one entry for each spike generated in the Simulated Motor Cortex and one entry for each instance of noise incorrectly detected as a spike. Each 32-bit wide entry will contain bits that identify the spike according to its electrode, source neuron, and its timestamp. Each entry will also have bits that indicate whether the spike has been detected and the results (if applicable) of spike sorting. The memory will need to be approximately 250,000 words deep, sufficient to handle a worst-case scenario of all 2,500 channels each firing 100 times in a single second. Memory requirements will be met by either the on-chip Block-RAM modules of the Xilinx Virtex 4 family or by a standalone external memory.

As the spike detector processes the neural waveforms, the detected events will be compared against the Spike Train Memory. When a detected event matches the timestamp and channel number of a spike in the Memory, the “detected” bit for that spike will be set. If no matching entry is found, the detected event is declared to be a false alarm and is stored into the Memory.

In a similar fashion, the spike sorter module will also annotate entries in the Spike Train Memory as detected events are sorted. In this manner, a complete history of the spike processing activity for each spike will be logged by the system. *This will create an unprecedented tool for analyzing spike processing in fine-grained detail with a large number of channels.* The Spike Train Memory will be periodically flushed into an “External Memory” hard drive in order to facilitate off-line analysis and to make room for newer spikes. The External Memory will also store both the kinematic stimuli that are encoded by the cortex and the stimulus estimate generated by the reconstruction block. These will be compared in an off-line analysis that will quantify the system’s overall ability to faithfully encode, communicate, and decode information. This can be accomplished using either a simple linear regression [2] or a more sophisticated computation of the mutual information rate [24,69].

Programmable logic devices have been shown to be well suited for neural processing and neural modeling tasks with large numbers of identically structured data channels [70-72]. Indeed, fine-grained programmable logic is the best choice for almost any application in which many signals must be similarly processed in parallel [73]. In such settings, conventional sequential processor computers can be substantially slower than massively parallel, pipelined programmable logic, even with the advantage of far greater clock speeds [74]. Programmable logic is also superior to conventional digital signal processing (DSP) in this context since, if needed, existing DSPs can be implemented as “soft cores” in programmable logic [73]. The PI’s collaborator, Dr. Dennis Silage, is a noted authority in large-scale massively parallel programmable logic devices and will work with the PI’s graduate students on all aspects related to design and testing of the srBMI. Of particular importance will be the software tools and hardware architectures for interconnecting the individual programmable logic devices, since it is unlikely that a single device will suffice for a task of the scope proposed here. Although various compilers and interconnection architectures have been proposed in this area, it is unclear how optimally suited they are for massively parallel designs. The PI, in collaboration with Dr. Silage, will work to establish which of the many existing partitioning and scheduling algorithms will optimize performance of the proposed massively parallel multi-device architecture [73,75-77].

A key advantage of the proposed system is that spike arrival times are known *exactly*, regardless of signal-to-noise ratio. Thus it will be possible to accurately quantify the performance of each of the spike processing blocks. This arrangement is superior to procedures for evaluating spike processing that depend on actual cortical recordings, since in those cases, spike times must be *estimated* by a user-supervised computer algorithm. This advantage, coupled with the fact that the system will be used in a closed-loop configuration in which the stimuli will be generated by an adaptive control loop, will result in a unique and powerful tool for rapidly, accurately, and flexibly modeling BMI instrumentation.

The entire design of Research Aim 2, including both hardware and software, will be made publicly available online in the hopes of standardizing the evaluation of BMI processing algorithms across laboratories; other investigators who test their own processing blocks using the srBMI will be encouraged to add their code to the public online repository.

D.3.4. Research Aim 3

The goal of Research Aim 3 will be to use the simulation system developed in Research Aims 1 and 2 to create a series of hypotheses about BMI instrumentation that will eventually be tested in animal models. Although the experiments proposed in this Aim are important in that they will lead to optimization strategies for BMI instrumentation, it is not feasible to conduct them in animal models. This is due in part to the sheer number of subjects that would be required to sufficiently evaluate all permutations of BMI configurations, and in part to the fact that it would be impossible to systematically vary spike processing parameters in an animal preparation. The

simulation platform developed under this proposal will be sufficiently realistic to identify candidate instrumentation configurations that are optimized for various applications. These will be testable in animal models, although that testing will be beyond the scope of the present research proposal.

The PI will seek to answer the following questions, none of which have ever been conclusively addressed in a closed-loop BMI context, simulated or otherwise: (1) How do spike detection errors (false alarms and missed detections) affect BMI performance (2) How do spike sorting errors (misclassifications) affect BMI performance (3) What is the effect on BMI performance when spike sorting is removed from the BMI signal processing pathway (4) What is the effect of spike latency on BMI performance (5) What is the effect of bit error rate on BMI performance? Although all of these questions have been addressed to some extent in open-loop simulations (spike detection [14,16,17,19,21,78-83]; spike sorting [2,15,21-25,84-86]; latency [33]; bit error rate [87]), the systematic effects of neural adaptation in overcoming system deficits have been omitted.

One of the most valuable aspects of the proposed model with respect to quantifying BMI performance is that the Simulated Motor Cortex (see Figure 4) creates both extracellular cortical electrode traces *and* the corresponding spike trains. This will allow BMI instrumentation to be evaluated in two distinct techniques. In the first technique, spike train processing is manipulated, allowing the PI to measure BMI behavior in response to precisely controlled instrumentation performance. For example, it will be possible to program the srBMI to randomly drop 10% of all spikes (in proportion to the instantaneous sampling rate) or to add “falsely detected” spikes at random. This type of analysis is not concerned with action potential waveforms or signal to noise ratios, and is therefore effectively a measure of the system’s ability to maximize instantaneous information throughput. In a similar vein, the PI will be able to systematically investigate how the BMI model responds when spikes from different neurons on the same electrode are misclassified at specific rates during spike sorting. The PI will also quantify the effects of eliminating spike sorting and thus pooling all spikes from the same electrode. This analysis is similar to the technique proposed by Won but with the added realism of error-mediated plasticity in the neural control mechanism [25].

In the second BMI evaluation technique, actual neural signal processing algorithms will be embodied in the srBMI. This will allow, amongst other experiments, the comparison of specific algorithms for spike sorting, automatic threshold computation, and spike detection. It will also allow investigation of the bit-error rate, a vital metric to characterize for wireless and implantable systems. In these cases, erroneous bits may result in non-sensical waveforms that can be misdetected or sorted. The srBMI will allow the random insertion of erroneous bits at various rates, effectively modeling a non-ideal wireless channel.

In all cases, the goal will be to investigate how BMI performance varies as a function of the parameter in question. The PI will investigate performance metrics at both steady state and during adaptation. BMI performance will be quantified using the one and two degree-of-freedom adaptations of Fitts’ Law as outlined in Research Aim 1, where “index-of-difficulty” (ID) is the standard metric; greater ID value indicates poorer overall prosthesis control [60].

Based on published neuron-dropping analyses [2], the PI hypothesizes that a general three-tiered performance scale will be observed in all experiments. In the first phase, the adaptive neural structures will be able to overcome small errors in spike processing, producing prosthesis control that is equivalent to the ideal, zero error case. In the second phase, medium-range errors will yield a linear decrease in prosthesis control. In the third phase, the performance decline is expected to effectively level off, such that increased error rates will no longer produce measurable performance loss.